



**NORTH SYDNEY GIRLS HIGH SCHOOL
YEAR 12 – TERM 2 ASSESSMENT**

2007

MATHEMATICS EXTENSION 1

TIME ALLOWED: 60 minutes
Plus 2 minutes reading time

INSTRUCTIONS:

- Start each question on a new page
- Hand each question in separately, including a sheet for non-attempts
- Show all necessary working

This task is worth 25% of the HSC Assessment Mark

QUESTION 1. (7 marks) Start a new page

Marks

a) Find the following indefinite integrals:

(i) $\int \frac{dx}{x^2}$ **1**

(ii) $\int (2x + 5)^4 dx$ **2**

b) Find the equation of the curve given $\frac{d^2y}{dx^2} = 2x - 1$. **4**

and when $x = 3$, $\frac{dy}{dx} = 8$, and the point $(0, -1)$ lies on the curve.

QUESTION 2. (8 marks) Start a new page

a) Convert $\frac{2\pi}{3}$ radians to degrees. **1**

b) Find the area of a sector of a circle of radius 8 cm if the length of the arc is 2 cm. **3**

c) Use Simpson's rule with 5 function values to approximate $\int_1^5 \frac{dx}{x+1}$. **4**

QUESTION 3. (8 marks) Start a new page

a) Differentiate with respect to x :

(i) e^{3x} **1**

(ii) $\ln 3x$ **1**

(iii) 3^x **2**

b) Differentiate $e^{2x} \cos x$ **2**

c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 4x - \sin x}{2x}$. **2**

QUESTION 4. (7 marks) Start a new page

a) (i) State the period and amplitude of $y = 3 \sin \frac{x}{2}$. **2**

(ii) Sketch the curve $y = 3 \sin \frac{x}{2}$ for $-\pi \leq x \leq \pi$. **2**

b) Solve $\tan 2x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$. **3**

continued on next page

QUESTION 5. (7 marks) Start a new page

Marks

- a) Find $\int_0^{\frac{\pi}{8}} \cos^2 2x \, dx$. **3**
- b) A person invests \$2000 at the beginning of each year in a superannuation fund. Assuming that interest is paid at the rate of 9% pa on this money,
- (i) show that at the end of n years the investment will have a value of $\frac{218\,000(1.09^n - 1)}{9}$ dollars. **2**
- (ii) find the number of full years required for the accumulated value to reach \$100 000. **2**

QUESTION 6. (8 marks) Start a new page

- a) Using the substitution $u = 2 + x^2$, find $\int 3x(2 + x^2)^3 \, dx$. **3**
- b) (i) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$. **2**
- (ii) Given that $y = (\sec x + \tan x)^m$ and m is a positive integer show that $\frac{dy}{dx} = my \sec x$. **3**

End of paper

Year 12 Term 2 2007 Extension 1

Question 1

a) $\int x^{-2} dx = -\frac{1}{x} + C$ (1)

ii) $\int (2x+5)^4 dx = \frac{(2x+5)^5}{10} + C$ (2)

b) $\frac{dy}{dx} = 2x - 1$

$\frac{dy}{dx} = x^2 - x + C$

$8 = 3^2 - 3 + C$

$C = 2$

$\frac{dy}{dx} = x^2 - x + 2$

$y = \frac{x^3}{3} - \frac{x^2}{2} + 2x + D$

$-1 = 0 + D$

$y = \frac{x^3}{3} - \frac{x^2}{2} + 2x - 1$ (4)

Question 2

a) $\frac{2\pi}{3} = \frac{2}{3} \times 180^\circ = 120^\circ$ (1)

b) $2 = 80 \rightarrow 0 = \frac{1}{4}$

Area = $\frac{1}{2} \times 8^2 \times \frac{1}{4}$

$= 8 \text{ cm}^2$ (3)

c) $\int_1^5 \frac{dx}{x+1} = \int_1^3 \frac{dx}{x+1} + \int_3^5 \frac{dx}{x+1}$

$= \frac{3-1}{6} \left(\frac{1}{2} + 4 \times \frac{1}{3} + \frac{1}{4} \right) + \frac{5-3}{6} \left(\frac{1}{4} + 4 \times \frac{1}{5} + \frac{1}{6} \right)$

$= 1.1$ (4)

Question 3

a) $\frac{d}{dx}(e^{3x}) = 3e^{3x}$ (1)

ii) $\frac{d}{dx}(3^x) = \ln 3 \times 3^x$ (1)

iii) $\frac{d}{dx}(\ln(2x-1) - \ln(3x+1)) = \frac{2}{2x-1} - \frac{3}{3x+1}$ (2)

b) $\frac{d}{dx}(e^{2x} \cos x) = 2e^{2x} \cos x - e^{2x} \sin x$ (2)

c) $\lim_{x \rightarrow 0} \left(\frac{\sin 4x}{2x} - \frac{\sin x}{2x} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin 4x \times 2}{4x} - \frac{\sin x \times \frac{1}{2}}{x} \right) = 2 - \frac{1}{2} = \frac{3}{2}$ (2)

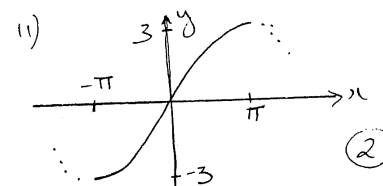
x	1	2	3	4	5
f(x)	1/2	1/3	1/4	1/5	1/6

Question 4

a) $y = 3 \sin \frac{x}{2}$
period = $\frac{2\pi}{(\frac{1}{2})}$

$= 4\pi$

amplitude = 3 (2)



b) $\tan 2x = -\sqrt{3}$ $0 \leq x \leq 2\pi$
 $0 \leq 2x \leq 4\pi$

$2x = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, 4\pi - \frac{\pi}{3}$

$= \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}, \frac{11\pi}{3}$

$\therefore x = \frac{\pi}{3}, \frac{5\pi}{6}, \frac{4\pi}{3}, \frac{11\pi}{6}$ (3)

Question 5

a) $\int_0^{\pi/8} \cos^2 2x dx = \frac{1}{2} \int_0^{\pi/8} (\cos 4x + 1) dx = \frac{1}{2} \left[\frac{1}{4} \sin 4x + x \right]_0^{\pi/8} = \frac{1}{2} \left(\frac{1}{4} \sin \frac{\pi}{2} + \frac{\pi}{8} - \left(\frac{1}{4} \sin 0 + 0 \right) \right) = \frac{1}{2} \left(\frac{1}{4} + \frac{\pi}{8} \right) = \frac{\pi + 2}{16}$ (3)

Question 5 cont

b) 1st \$2000 invested for n yrs $2000(1.09)^n$
 2nd ————— $n-1$ — $2000(1.09)^{n-1}$
 ⋮
 nth ————— 1 — $2000(1.09)$
 Total = $2000(1.09 + 1.09^2 + \dots + 1.09^n)$
 GP

∴ after n years total value is

$$\frac{2000 \times 1.09(1.09^n - 1)}{1.09 - 1}$$

$$= \frac{2000(1.09)(1.09^n - 1)}{0.09}$$

$$= \frac{218000(1.09^n - 1)}{9} \quad (2)$$

ii) $\frac{218000}{9}(1.09^n - 1) = 100000$

$$1.09^n - 1 = 100000 \times \frac{9}{218000}$$

$$1.09^n = \frac{900}{218} + 1$$

$$= \frac{1118}{218}$$

$$n \log 1.09 = \log \left(\frac{1118}{218} \right)$$

$$n = \frac{\log \left(\frac{1118}{218} \right)}{\log(1.09)}$$

$$= 18.970 \dots$$

∴ 19 full years are required

(2)

Question 6

a) $\int 3x(2+x^2)^3 dx$

$$u = 2+x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$\therefore x dx = \frac{1}{2} du$$

$$\int 3 \times u^3 \times \frac{1}{2} du$$

$$= \frac{3}{2} \frac{u^4}{4} + C$$

$$= \frac{3}{8} (2+x^2)^4 + C \quad (3)$$

b) $\frac{d}{dx}(\sec x)$

$$= \frac{d}{dx}(\cos x)^{-1}$$

$$= -\frac{1}{\cos^2 x} \times (-\sin x)$$

$$= \frac{\sin x}{\cos^2 x} \times \frac{1}{\cos x}$$

$$= \tan x \sec x \quad (2)$$

ii) $y = (\sec x + \tan x)^m$

$$\frac{dy}{dx} = m(\sec x + \tan x)^{m-1} (\sec x \tan x + \sec^2 x)$$

$$= m \sec x (\sec x + \tan x)^{m-1} (\sec x + \tan x)$$

$$= m \sec x (\sec x + \tan x)^m$$

$$= m \sec x \times y$$

$$= m y \sec x$$

(3)

Question 5

(a) $\cos 2A = 2\cos^2 A - 1 \Rightarrow 2\cos^2 A = 1 + \cos 2A$

$$\begin{aligned}\int_0^{\frac{\pi}{8}} \cos^2 2x dx &= \frac{1}{2} \int_0^{\frac{\pi}{8}} 2\cos^2 2x dx = \frac{1}{2} \int_0^{\frac{\pi}{8}} (1 + \cos 4x) dx \\&= \frac{1}{2} \left[x + \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{8}} \\&= \frac{1}{2} \left(\frac{\pi}{8} + \frac{1}{4} \sin \frac{\pi}{2} \right) - 0 \\&= \frac{\pi}{16} + \frac{1}{8}\end{aligned}$$

(b) (i) The first \$2000 will accumulate to $2000(1.09)^n$.

The second \$2000 will accumulate to $2000(1.09)^{n-1}$ and so on until the last \$2000 will accumulate to $2000(1.09)$

The investment will have an accumulated value of \$A, given by:

$$A = 2000(1.09)^n + 2000(1.09)^{n-1} + \dots + 2000(1.09)$$

$$A = 2000(1.09)^n + 2000(1.09)^{n-1} + \dots + 2000(1.09)$$

$$= 2000 \left[\underbrace{1.09 + \dots + (1.09)^{n-1} + (1.09)^n}_{a=1.09, r=1.09} \right]$$

$$= 2000 \times 1.09 \times \left(\frac{1.09^n - 1}{0.09} \right)$$

$$= \frac{218\,000(1.09^n - 1)}{9}$$

Alternatively

Let A_n be the amount accumulated after n years.

$$A_1 = 2000 \times 1.09$$

$$A_2 = (A_1 + 2000)1.09 = 2000(1.09)^2 + 2000(1.09)$$

$$A_3 = (A_2 + 2000)1.09 = 2000[1.09 + 1.09^2 + 1.09^3]$$

$$\therefore A_n = (A_{n-1} + 2000)1.09 = 2000[1.09 + 1.09^2 + 1.09^3 + \dots + 1.09^n]$$

$$A_n = 2000 \left[\underbrace{1.09 + \dots + (1.09)^{n-1} + (1.09)^n}_{a=1.09, r=1.09} \right]$$

$$= 2000 \times 1.09 \times \left(\frac{1.09^n - 1}{0.09} \right)$$

$$= \frac{218\,000(1.09^n - 1)}{9}$$

$$(ii) \quad A = \frac{218\,000(1 \cdot 09^n - 1)}{9} = 100\,000$$

$$\therefore 1 \cdot 09^n - 1 = \frac{100\,000 \times 9}{218\,000} = \frac{450}{109}$$

$$\therefore 1 \cdot 09^n = \frac{559}{109}$$

$$\therefore \ln(1 \cdot 09^n) = \ln\left(\frac{559}{109}\right)$$

$$\therefore n \ln 1 \cdot 09 = \ln\left(\frac{559}{109}\right)$$

$$\therefore n = \ln\left(\frac{559}{109}\right) \div \ln 1 \cdot 09$$

$$\therefore n \approx 18 \cdot 970$$

So 19 years will be needed for the accumulated value to reach \$100 000